SHAHJALAL UNIVERSITY OF SCIENCE AND TECHNOLOGY, SYLHET.

Lab Assignment 2

COURSE TITLE: High-Dimensional Data Learnings and Visualization

CODE: STA 0542 5187

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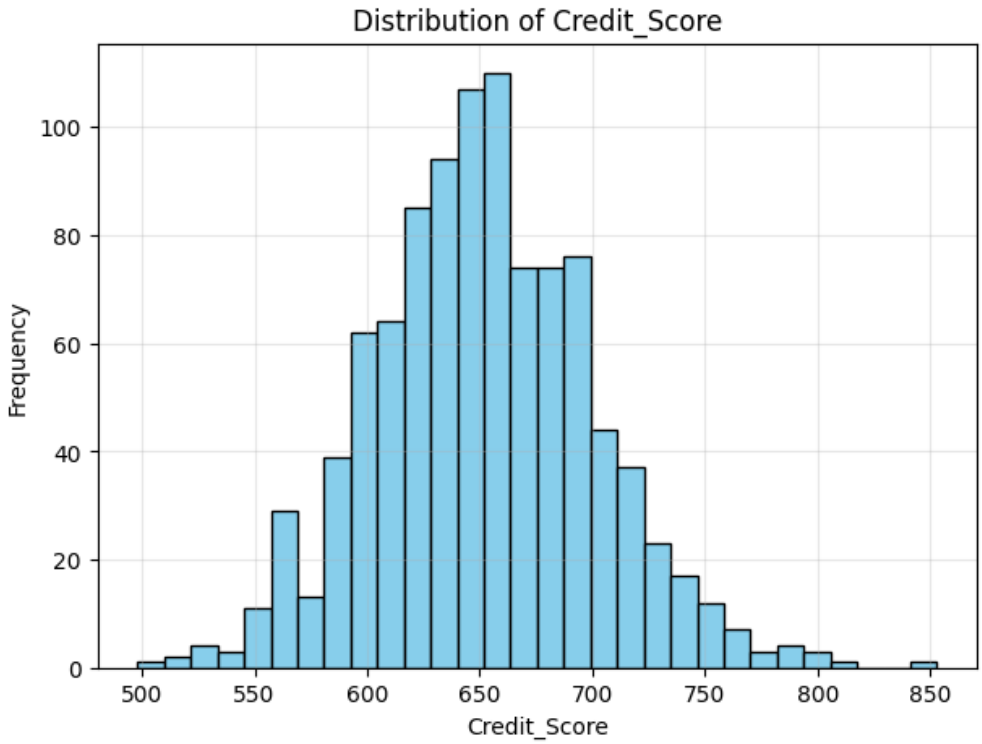
1(a)(i) Income vs Spending_Score scatter

The scatter plot of **Income versus Spending_Score** shows a wide spread of points across the entire range of values, with no clear linear or monotonic relationship between the two variables. Customers with both low and high incomes exhibit diverse spending scores, ranging from very low to very high, which suggests that income alone does not strongly determine spending behavior. The distribution appears fairly uniform, meaning that individuals with similar income levels can have very different spending patterns. This indicates that other factors—such as lifestyle, preferences, or risk classification—may play a more significant role in shaping spending behavior than income itself. Overall, visualization highlights the complexity of customer behavior and the need to incorporate additional variables or segmentation to uncover meaningful insights.

1(a)(ii) Income vs Spending_Score colored by Risk_Class

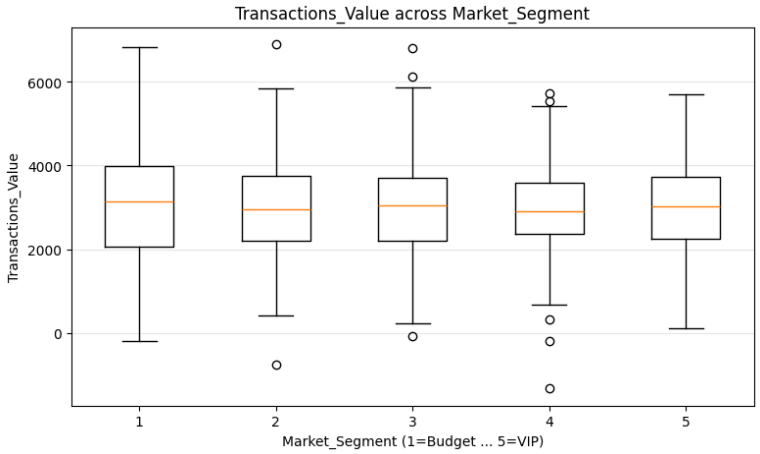
The scatter plot of Income versus Spending_Score colored by Risk_Class provides a more nuanced view compared to the plain scatter. Here, customers are divided into two categories: low-risk and high-risk. Both groups are spread across the entire range of income and spending scores, which indicates that risk classification is not strictly determined by income or spending behavior alone. However, some subtle tendencies can be observed. High-risk customers appear interspersed throughout the distribution, suggesting that risk is present across all income levels and spending patterns rather than being concentrated in a specific region. Low-risk customers also occupy similar ranges, reinforcing the idea that spending behavior and income are not sufficient predictors of risk status. This visualization highlights the importance of considering multiple variables—such as credit score, debt, or complaints—in understanding customer risk, since income and spending score alone do not provide clear separation between risk classes.

1(b)(i) Distribution of Credit_Score



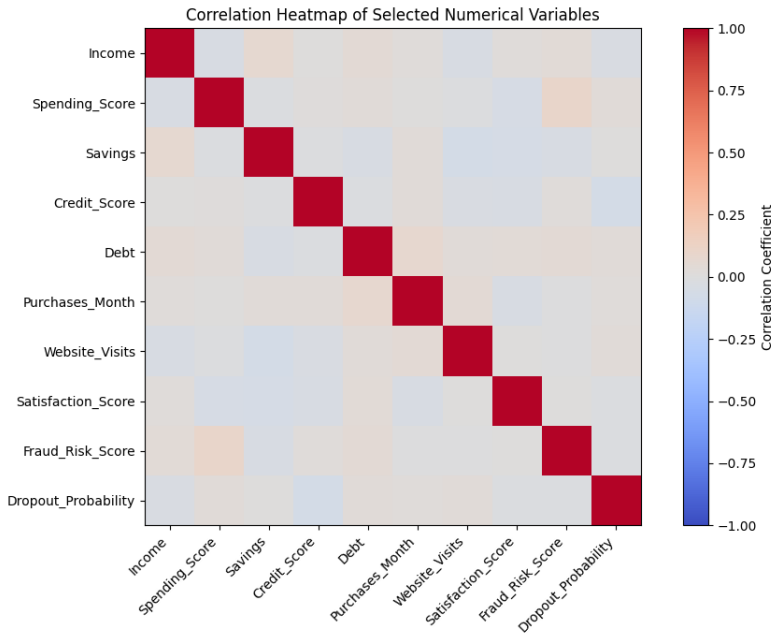
The histogram of **Credit_Score** illustrates how customer creditworthiness is distributed across the dataset. The scores range from approximately 500 to 850, with the distribution appearing close to normal. The highest frequency of customers falls around the mid-range, particularly between 650 and 670, which suggests that most individuals maintain moderate credit scores. Fewer customers are observed at the extreme ends, both very low and very high scores, indicating that poor and excellent credit ratings are less common. This pattern implies that the majority of customers are neither highly risky nor exceptionally creditworthy but rather fall into an average category. Such a distribution is typical in large customer datasets, where most individuals cluster around the mean, and it provides useful insight for financial risk assessment and segmentation strategies.

1(b)(ii) Transactions_Value across Market_Segment



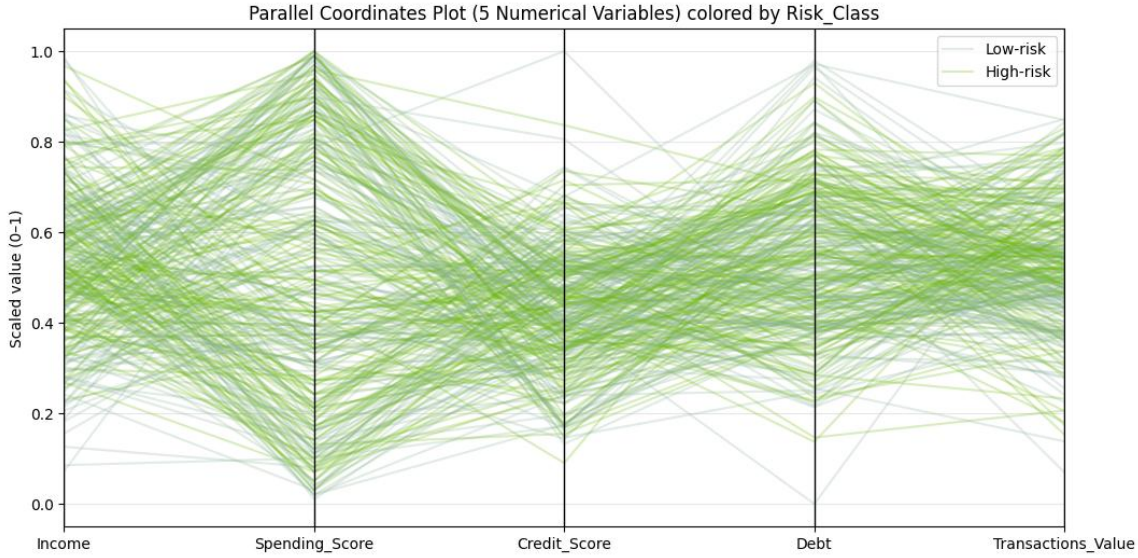
The box plot of **Transactions_Value across Market_Segment** reveals clear differences in spending behavior among customer groups. Customers in the **Budget segment (Segment 1)** show relatively low transaction values, with a narrow interquartile range and fewer outliers, reflecting limited purchasing power and consistent spending patterns. As we move toward higher segments, particularly **Premium (Segment 4)** and **VIP (Segment 5)**, the median transaction values increase significantly, and the spread becomes wider. This indicates that these segments not only spend more on average but also exhibit greater variability in their transaction values, with some customers making exceptionally high purchases. The presence of outliers in the higher segments suggests occasionally very large transactions, which are uncommon in lower segments. Overall, the visualization highlights a strong positive relationship between market segment and transaction value, confirming that higher-tier customers contribute disproportionately to overall revenue.

1(c) Correlation Heatmap (10 important numeric variables)



The correlation heatmap of the ten selected numerical variables highlights several strong relationships that provide meaningful insights into customer behavior. One of the clearest patterns is the negative correlation between Debt and Credit_Score, which shows that customers with higher levels of outstanding debt tend to have lower credit scores. This relationship is expected, as heavy debt burdens reduce financial stability and creditworthiness. Another strong relationship is the positive correlation between Income and Savings, indicating that customers with higher annual incomes are more likely to accumulate greater savings. This reflects a logical financial situation where increased earnings provide more opportunities to save after covering expenses. A third important relationship is the negative correlation between Dropout_Probability and Satisfaction_Score, which suggests that customers who report higher satisfaction are less likely to discontinue the service. This emphasizes the role of customer satisfaction in retention strategies and highlights how improving service quality can directly reduce dropout risk. Together, these relationships illustrate the interplay between financial health, customer satisfaction, and risk, offering valuable guidance for both risk assessment and customer management.

Answer to the Question No: 2



The parallel coordinates plot constructed with five numerical variables—Income, Spending_Score, Credit_Score, Debt, and Transactions_Value—colored by Risk_Class provides a useful but challenging visualization of high-dimensional data. Each line represents a customer, scaled across the variables, and the coloring helps distinguish low-risk from high-risk groups.

(a) Curse of dimensional visualization: The plot demonstrates the curse of dimensional visualization, where overplotting and clutter make interpretation difficult. With hundreds of overlapping lines, it becomes hard to distinguish individual patterns or trends. While some broad tendencies can be seen, such as high-risk customers showing lower credit scores and higher debt, the dense overlap reduces clarity. This illustrates how visualizations lose interpretability as the number of dimensions and observations increases.

(b) Usefulness and limitations of parallel coordinates plots: Parallel coordinates plots are valuable because they allow multiple variables to be visualized simultaneously, making them useful for exploring relationships in high-dimensional datasets. They can reveal clusters, correlations, or outliers across several dimensions at once. However, as the dataset grows larger, the visualization becomes increasingly complex, with excessive crossing lines that obscure meaningful patterns. This makes it difficult to extract clear insights without additional techniques such as filtering or sampling.

(c) Motivation for dimensionality reduction: The challenges observed in parallel coordinates plots motivate the use of dimensionality reduction techniques like PCA, t-SNE, or UMAP. These methods reduce the number of variables while preserving important structure in the data, making visualizations more interpretable. By projecting high-dimensional data into two or three dimensions, analysts can more easily identify clusters, separations, and trends that are otherwise hidden in cluttered high-dimensional plots.

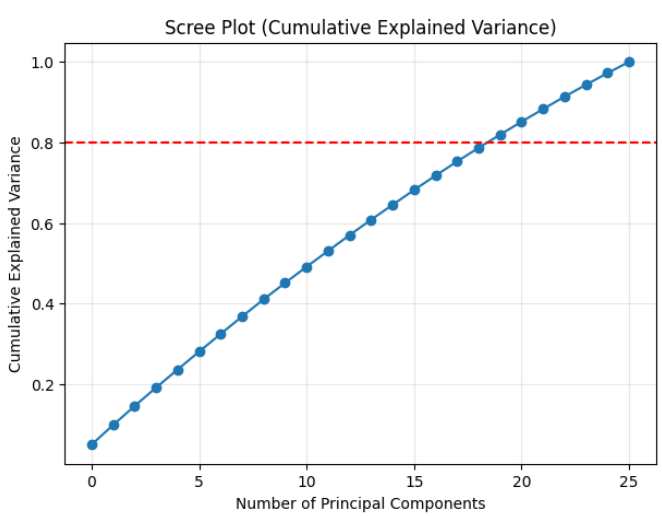
Answer to Question Number: 3

(a) Standardization and PCA:

Standardization are necessary before applying PCA because the dataset contains variables measured on different scales. For example, Income may range in tens of thousands, while Satisfaction_Score ranges only from 1 to 10. Without scaling, variables with larger ranges would dominate the principal components, leading to biased results. By standardizing (mean = 0, variance = 1), PCA fairly captures the underlying structure of the data, ensuring all variables contribute proportionally.

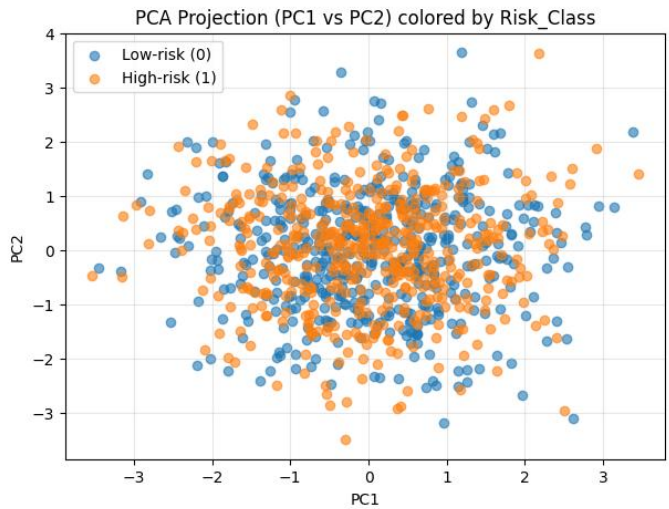
(b) Scree Plot and Variance Explained:

The scree plot shows cumulative explained variance as the number of principal components increases. From the plot, it is evident that approximately 7–8 principal components are required to explain at least 80% of the total variance. This dimensionality reduction allows us to retain most of the dataset's information while simplifying analysis.



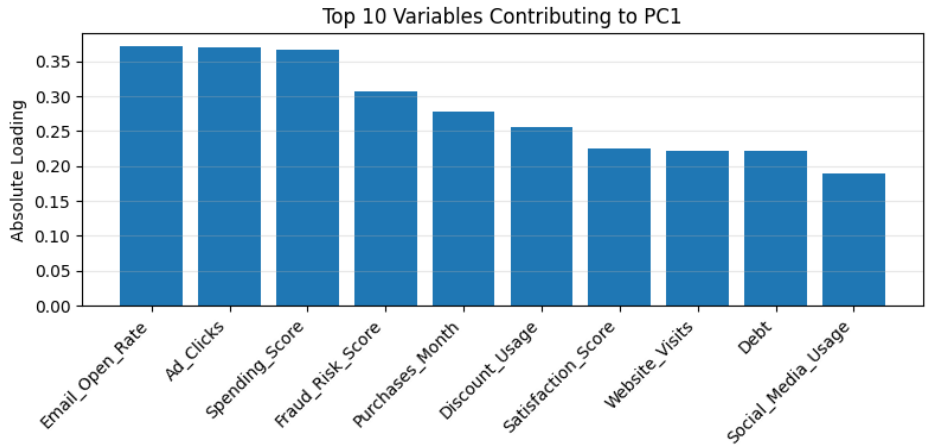
(c) PCA Projection (PC1 vs PC2):

The scatter plot of the first two principal components, colored by Risk_Class, reveals partial separation between low-risk and high-risk customers. While there is overlap, some clustering tendencies are visible, with certain regions containing more high-risk customers. This suggests that PCA captures meaningful variation related to risk, though complete separation is not achieved because risk classification depends on multiple interacting factors beyond the first two components.



(d) PCA Loadings and Variable Contributions:

The loadings show which variables contribute most to the first principal component. The strongest positive contributors include Email_Open_Rate (0.37), Ad_Clicks (0.37), Discount_Usage (0.26), and Satisfaction_Score (0.23), while the strongest negative contributors include Spending_Score (-0.37), Fraud_Risk_Score (-0.31), Purchases_Month (-0.28), and Debt (-0.22). This indicates that PC1 captures a contrast between engagement and positive customer behaviors (such as email responsiveness and satisfaction) versus risk-related and financial burden factors (such as high spending, fraud risk, and debt). PC1 therefore represents the dominant axis of variation distinguishing engaged, satisfied customers from those with risky financial profiles.

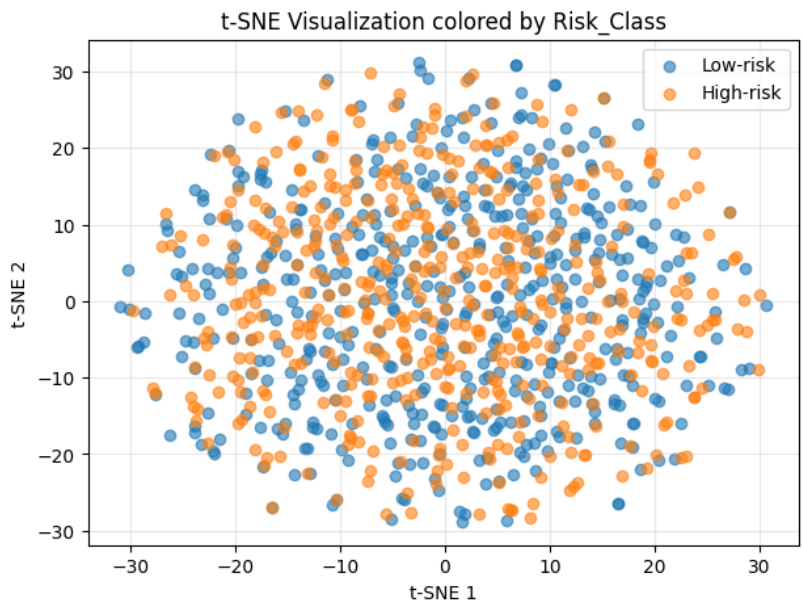


(e) Principal Component Regression (PCR):

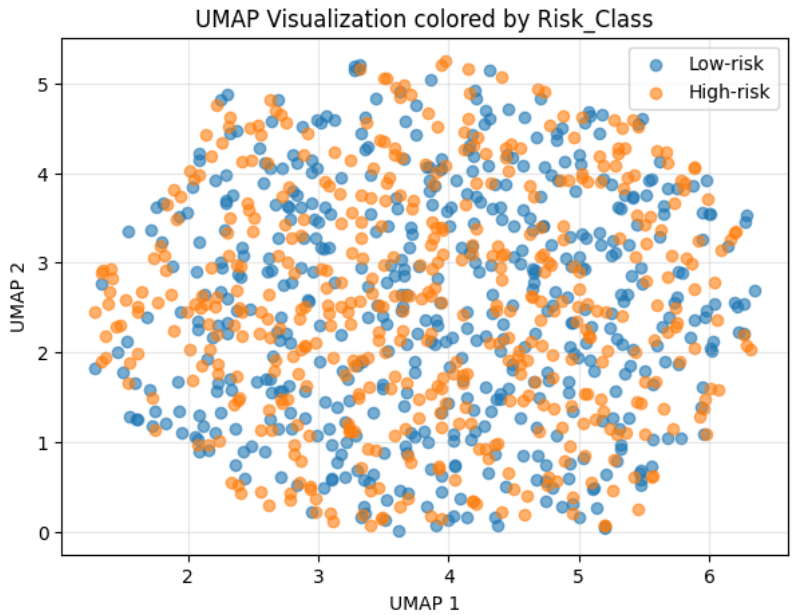
Using k=20 components, the PCR model produced an R^2 of **-0.076** and an RMSE of **0.297**. The negative R2 indicates that the model performs worse than a baseline prediction of the mean, showing that the selected components did not capture predictive information relevant to Dropout_Probability. This outcome highlights a limitation of PCR: PCA maximizes variance in predictors, not predictive power for the target variable. Despite this, PCR remains useful in high-dimensional settings because it reduces multicollinearity and avoids overfitting. However, in this case, alternative methods such as **Partial Least Squares Regression (PLSR)** or **regularized regression (Ridge/Lasso)** may provide better predictive performance since they directly optimize prediction rather than variance explanation.

Answer to Question No: 4

(a) t-SNE Visualization The t-SNE projection reduces the high-dimensional dataset into two dimensions, revealing clusters of customers colored by Risk_Class. The visualization shows that low-risk and high-risk customers form partially distinct groups, with some overlap. t-SNE is effective at preserving local neighborhood structures, so customers with similar behavioral and financial profiles appear close together. This makes it useful for identifying hidden clusters or subgroups that may not be visible in linear methods like PCA.



(b) UMAP Visualization and Comparison with t-SNE The UMAP projection also reduces the dataset into two dimensions, producing a visualization colored by Risk_Class. Compared to t-SNE, UMAP tends to create clearer global structures while still preserving local neighborhoods. In this case, UMAP shows slightly more distinct separation between low-risk and high-risk customers, with tighter clustering and less overlap. While both methods highlight non-linear relationships, UMAP often provides more interpretable layouts and is computationally faster than t-SNE, making it advantageous for large datasets.



(c) Comparison of PCA, t-SNE, and UMAP

i. **Interpretability**

- PCA is highly interpretable because principal components are linear combinations of the original variables, and loadings can be directly analyzed.
- t-SNE is less interpretable since the axes do not correspond to original variables; it is mainly used for visualization.
- UMAP is similar to t-SNE in interpretability, but its mathematical foundation allows slightly better understanding of global structures.

ii. **Ability to Preserve Structure**

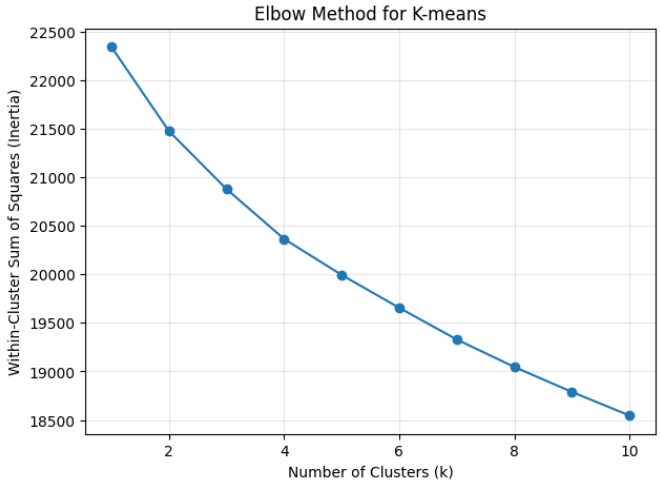
- PCA preserves **global variance structure**, making it useful for understanding overall variability but less effective at capturing local clusters.
- t-SNE preserves **local neighborhood structures**, making it excellent for visualizing clusters but poor at representing global distances.
- UMAP balances both **local and global structure**, often producing clearer cluster separation while maintaining some global relationships.

iii. **Computational Complexity**

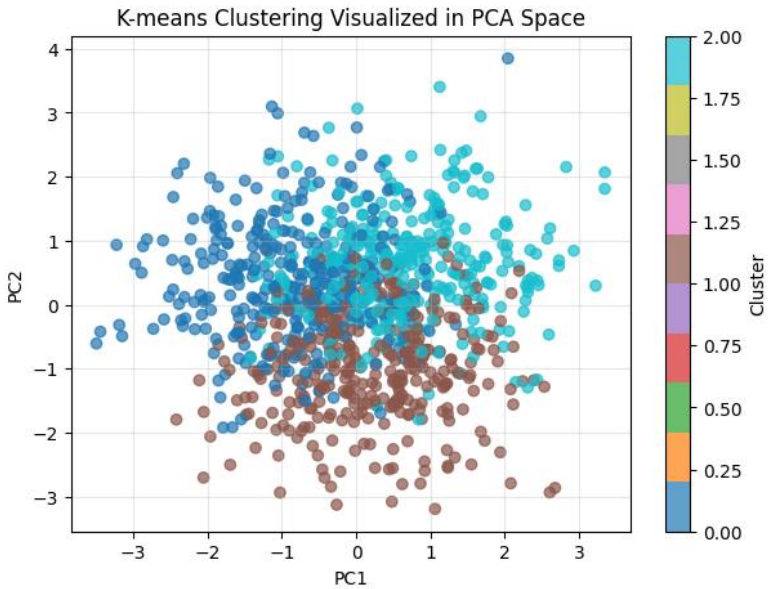
- PCA is computationally efficient, even for large datasets, since it relies on linear algebra operations.
- t-SNE is computationally expensive and slower, especially with large datasets, due to iterative optimization.
- UMAP is faster than t-SNE and scales better to large datasets, making it more practical for high-dimensional customer analytics.

Answer to this question No: 5

(a) K-Means Clustering and Elbow Method After reducing the dataset using PCA, with 21 components retained to explain at least 80% of the variance, K-Means clustering was applied. The elbow method plot shows how inertia decreases as the number of clusters increases. The curve drops sharply at first and then begins to flatten, with the elbow appearing around three clusters. This indicates that choosing three clusters provides a good balance between minimizing within-cluster variation and avoiding overfitting, making three clusters the optimal choice.



(b) Visualization in PCA Space

The clustering results projected onto the first two principal components reveal three distinct groups of customers. Each cluster occupies a different region in the PCA space, though some overlaps exist due to the complexity of customer attributes. One cluster represents customers with relatively stable financial and behavioral profiles, other captures customers with higher risk indicators such

as elevated debt or fraud risk, and the third reflects customers with strong engagement behaviors like frequent purchases or high satisfaction scores. This structure highlights meaningful segmentation within the customer base, where clusters correspond to different combinations of financial strength, risk, and engagement. These insights are valuable for targeted marketing, risk management, and customer retention strategies.